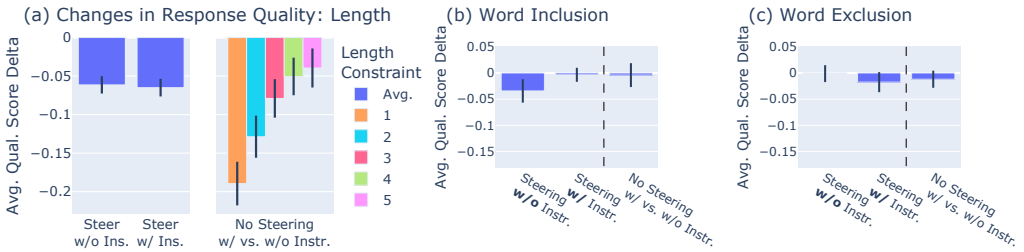

Figure 9: **Changes in the Response Quality Scores.** Quality score deltas for (a) length instructions, (b) word inclusion, and (c) word exclusion, across three conditions: steering without instructions, steering with instructions, and no steering with vs. without instructions.

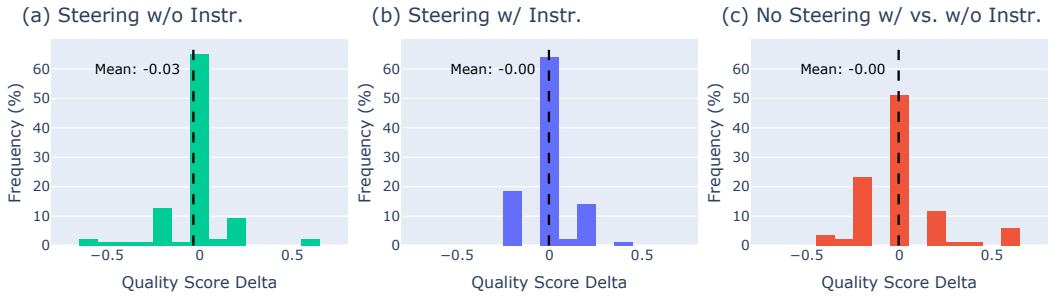

Figure 10: **Distribution of Quality Score Changes for Word Inclusion.** Histograms of quality score deltas under three conditions: (a) steering without explicit instructions, (b) steering with explicit instructions, and (c) no steering, comparing outputs with and without instructions. The distributions show minimal impact on quality scores across all settings, with mean values close to zero, with most shifts falling within the [-0.25, 0.25] range.

Word-specific Instructions. In Figures 9b and 9c, we report the changes in quality scores for instructions that request the inclusion or exclusion of a specific word in the answer.

The decrease in quality is minimal across all scenarios, the largest drop observed when steering for word inclusion without explicit instructions (around -0.03). We further investigate this by examining the empirical distributions of the score deltas for word inclusion instructions (Figure 10). The distributions appear centered around 0, with a few outliers showing deviations larger than 0.4. We perform a paired two-sided t-test to assess whether the means of these three distributions are significantly different. The p-values for the three tests (one for each pair) are all greater than 0.05 (0.25, 0.98, and 0.26). While the lack of significance does not prove that the distributions are identical, we interpret this as evidence of the minimal impact our steering procedure has on the model's ability to address the base queries.

Format: Breakdown by Instruction. Figure 11 presents the quality score deltas for format instructions, broken down by individual instructions across all four models (Phi-3, Gemma 2 2B IT, Mistral 7B IT, and Gemma 2 9B IT). The three settings compared are steering without explicit instructions, steering with explicit instructions, and no steering but comparing outputs with and without explicit instructions.

F.4 Quality Score/Accuracy Trade-off

We investigate how the steering weight c mediates the trade-off between instruction-following accuracy and output quality degradation for word exclusion constraints. Figure 12 shows the quality score and accuracy obtained by steering Phi-3 with varying negative values of c (as we subtract the word-inclusion vector, as explained in §5 and Appendix J). In both settings (with and without explicit text instructions), larger absolute steering weights lead to higher instruction-following accuracy. This improvement comes at the cost of a gradual decrease in quality score, indicating a trade-off between strict adherence to constraints and maintaining output comprehensiveness. The

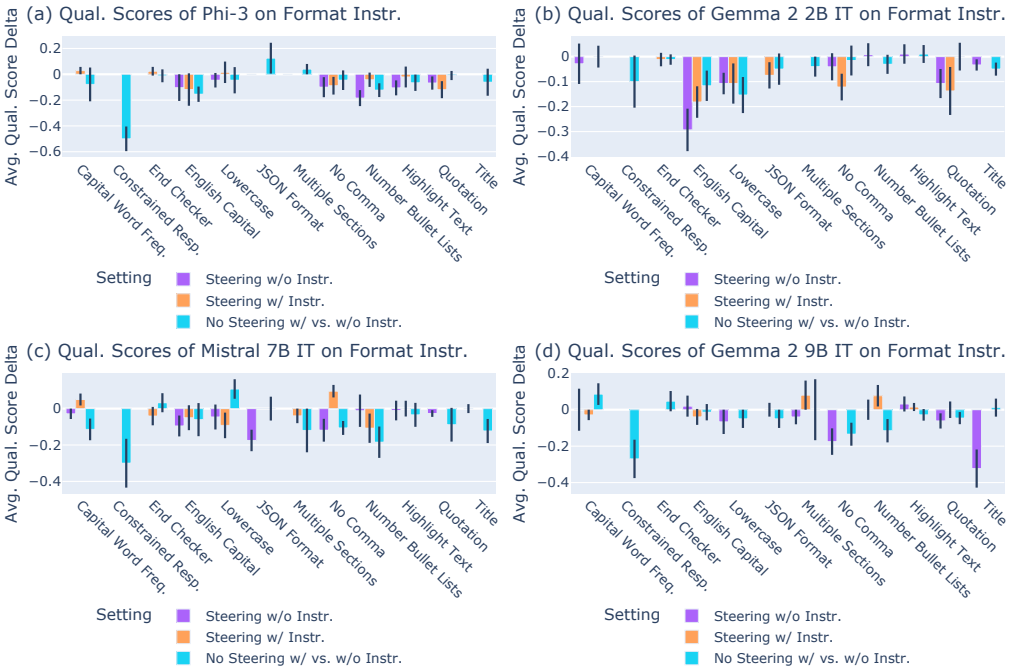

Figure 11: **Quality score deltas for format instructions across all models and settings.** (a) Phi-3, (b) Gemma 2 2B IT, (c) Mistral 7B IT, (d) Gemma 2 9B IT. Bars represent the average quality score delta under three conditions: steering without explicit instructions (purple), steering with explicit instructions (orange), and no steering but comparing outputs with and without explicit instructions (light blue).

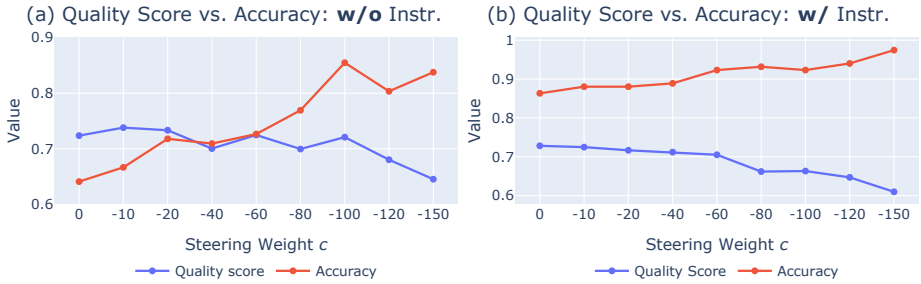

Figure 12: **Trade-off Between Quality Score and Accuracy on Word Exclusion.** (a) Results for steering without explicit text instructions, showing increasing accuracy at the cost of a gradual decrease in quality. (b) Results for steering with explicit instructions, demonstrating a similar trend with higher baseline accuracy.

smooth transitions in both metrics suggest that tuning c offers fine-grained control over this balance, highlighting the flexibility of activation steering.

G Implementation and Experimental Details

Tokenization and Decoding. Steering vectors are computed at the last token of the input. For instruction-tuned models, this typically corresponds to the `<|assistant|>` token, which marks the transition from user input to the start of the model's generation.[8] For non-instruction-tuned models, we follow previous work (Kojima et al., 2022; Yang et al., 2024) and structure the prompt as "*Q:* {`problem`}\n*A:*." Model outputs are decoded greedily, with a maximum generation length

[8] `https://huggingface.co/docs/transformers/main/en/chat_templating`

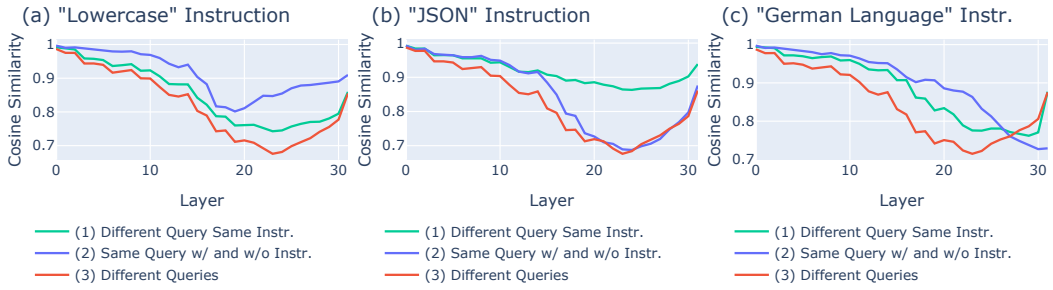

Figure 13: **Residual stream similarity across layers.** Cosine similarity of residual stream activations across layers for three instructions: (a) "Lowercase," (b) "JSON," and (c) "German Language." Comparisons are made between: (1) different queries with the same instruction (green), (2) same query with and without the instruction (blue), and (3) different queries without instruction (red). Results are shown for Phi-3 across all layers.

of 2048 tokens for format and length experiments, and 1024 tokens for keyword experiments. For efficiency, validation runs use a reduced maximum length of 384 tokens.

Tools and Libraries. For length-related experiments, word counts are measured as the number of space-separated sequences of characters and sentence counts are determined using NLTK (Bird & Loper, 2004). To assess whether the score improvement from steering is significantly different from standard inference, we carry out McNemar's test (McNemar, 1947). In particular, we use the exact version of the test, which uses the binomial distribution and is more conservative. The error bars reported in Figures 4 and 9 represent the standard error of the quality score computed over three different runs of GPT-4o. The smoothed empirical distributions shown in Figures Figure 5a, Figure 5c, Figure 8c, Figure 17a, and Figure 17c are obtained using kernel density estimation (Parzen, 1962). Our experiments were carried out using `PyTorch` (Paszke et al., 2019) and the `TransformersLens` library (Nanda & Bloom, 2022). We performed our data analysis using `NumPy` (Harris et al., 2020) and `Pandas` (Wes McKinney, 2010). Our figures were made using `Plotly` (Plotly Technologies Inc., 2015). The paper's bibliography was curated using `Ryanize-bib` (Zouhar, 2023).

H Additional Results: Representations

Similarity of Activations. Figure 13 shows the cosine similarity of residual stream activations in Phi-3 across different input sets. As described in §3, we compare the similarity of representations between: (1) pairs of inputs sharing the same base query, one with and one without the instruction; (2) inputs with different base queries but the same instruction; and (3) inputs with different base queries and no instruction. The similarity scores are reported across Phi-3's layers for three instructions: "lowercase" (no uppercase characters, Figure 13a), "JSON" (response formatted as JSON, Figure 13b), and "German Language" (Figure 13c).

Geometry of Steering Vectors. We perform a t-SNE dimensionality reduction on per-example instruction vectors, calculated as the difference in activations between inputs with and without explicit instructions, at layer 20 of Phi-3. The visualization for format instructions shows varying levels of cluster separation (Figure 14a). Specifically, instructions like "No Comma," "Lowercase," "JSON Format," "Title," and "Quotation" form distinct and well-separated clusters, indicating the model's ability to encode these constraints clearly. Instructions such as "Capitalize" and "Capital Word Frequency" cluster closely together, reflecting their similar semantic nature. In contrast, vectors for instructions like "End Checker" and "Constrained Response" are more dispersed, with no clear clustering, suggesting these instructions are less distinctly represented in the model's activation space.

Figure 14b presents a t-SNE dimensionality reduction of vectors computed for length instructions at layer 12 of Phi-3. Specifically, we include vectors derived from sentence-specific instructions (e.g., "answer using n sentences," $n \in 1, \ldots, 5$) as well as more general "conciseness" and "verbosity" instructions, as described in §4. A clear linear trend emerges, with vectors for longer output in-